background activity R_b, which has to be subtracted from the measured value R_m to obtain the true value for the radiation (e.g. from ^{32}P): $R = R_m - R_b$. The extrapolation of R to a value equal to R_b (i.e. C_3) gives the range.

Whereas it is possible to specify maximum ranges for charged particles, this is not possible for neutral particles such as neutrons and γ-quanta. If the absorber is not too thick, these particles undergo only one collision, or at the most a few, before they are absorbed. As a result the absorption curve has an exponential form.

$$\varphi = \varphi_0 e^{-\mu x} \tag{7.8}$$

where μ is the *total attenuation coefficient*. Thus for n and γ we have

$$\psi_{abs}(x) = e^{-\mu x} \tag{7.9}$$

The reduction in intensity of a beam can occur by two mechanisms. One involves the deflection or scattering of the particles from the direct line of path between the source and the detector and is described by the *scattering coefficient* μ_s. The second mode of reduction is the complete transfer of the projectile energy to the absorbing material (the particles are "captured") and is designated by the (*energy*) *absorption coefficient* μ_a. The (total) attenuation coefficient in Eqn (7.8) is the sum of both these mechanisms.

$$\mu = \mu_s + \mu_a \tag{7.10}$$

Both μ_s and μ_a can be measured independently. The (total) attenuation coefficient is of primary interest in radiation shielding, while the (energy) absorption coefficient is important in considering the effect of radiation on matter.

7.3. ABSORPTION OF PROTONS AND HEAVIER IONS

The mode of interaction of protons and heavier charged particles with the atoms of the absorbing material can be illustrated by considering the absorption of α-particles. With rare exception, α-particles emitted by radioactive nuclides have energies between 4 and 9 MeV. Since the α-particles are so much heavier than electrons, they are deflected very slightly when their Coulomb fields interact with atoms or molecules to form ion pairs. As a result, α-particles travel in a straight line as they pass through matter, which explains the straight paths observed for α-particles in cloud chamber photographs (Fig. 7.5). This is in contrast to the very curved or irregular paths of the secondary electrons emitted in the formation of the ion pair. For a 5 MeV α-particle the maximum energy of the secondary electrons is 2.7 keV. However only a small fraction of the secondary electrons actually receive this much energy; the average energy of the secondary electrons is closer to 100 eV. The ionization caused by more energetic secondary electrons is usually referred to as *δ-tracks* (cf. §8.2).

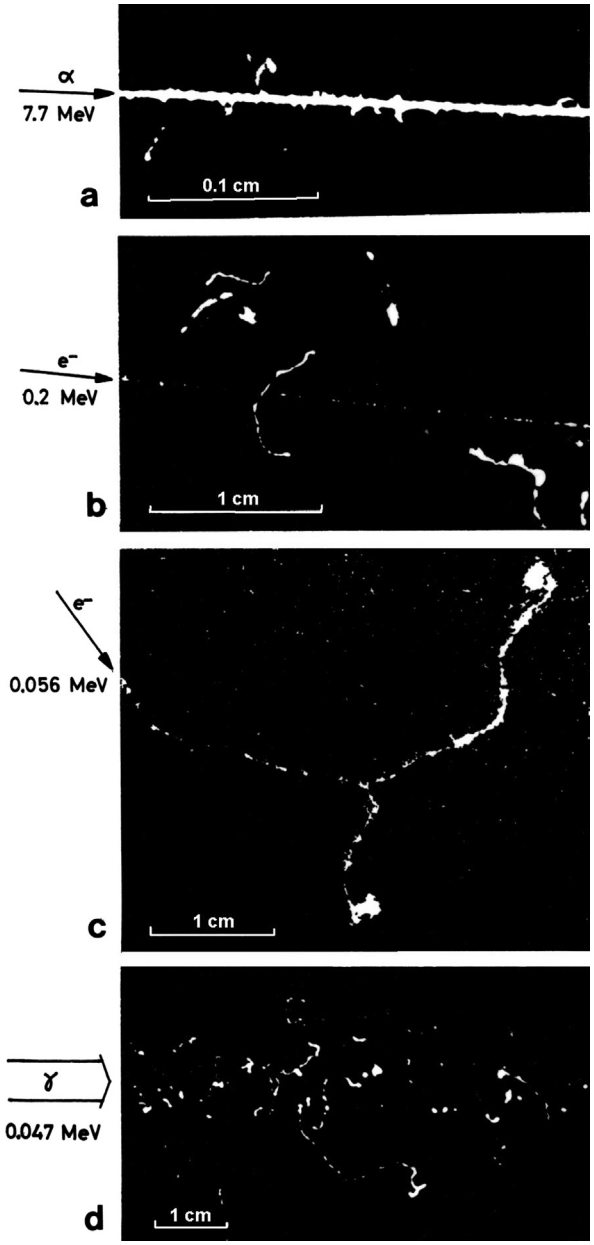


Figure 7.5 Cloud chamber tracks of α, β, (e^-), and γ-rays at 1 bar in air ((a), (b), and (c)) and in methane (d). *(From W. Gentner, H. Maier-Leibnitz, and H. Bothe.)*

Table 7.2 Range in water, and average linear energy transfer (LET) values for different radiation

Upper Half Refers to Monoenergetic (Accelerated) Particles. For β-decay $E_{abs} = 1/3\ E_{max}$

Radiation	Energy (MeV)	Maximum range		Average LET value in water (keV/μm)
		cm air	mm water	
Electron	1	405	4.1	0.24
	3	1400	15	0.20
	10	4200	52	0.19
Proton	1	2.3	0.023	43
	3	14	0.014	21
	10	115	1.2	8.3
Deuteron	1	1.7	—	—
	3	8.8	0.088	34
	10	68	0.72	14
Helium	1	0.57	0.0053	190
	3	1.7	0.017	180
	10	10.5	0.11	92
Fiss. fragment	100	2.5	0.025	3300
^{226}Ra (α)	E_α 4.80	3.3	0.033	145
^{210}Po (α)	E_α 5.30	3.8	0.039	136
^{222}Rn (α)	E_α 5.49	4.0	0.041	134
^{3}T (β)	E_{max} 0.018	0.65	0.0055	1.1
^{35}S (β)	E_{max} 0.167	31	0.32	0.17
^{90}Sr (β)	E_{max} 0.544	185	1.8	0.10
^{32}P (β)	E_{max} 1.71	770	7.9	0.07
^{90}Y (β)	E_{max} 2.25	1020	11	0.07
^{137}Cs (γ)	E_γ 0.66	$x_{½} = 8.1$ cm H_2O		0.39
^{60}Co (γ)	E_γ 1.20−1.30	$x_{½} = 11.1$ cm H_2O		0.27

In solids and liquids the total path length for α-particles from radioactive decay is quite short, only some micrometers. However, in gases at standard temperature and pressure the paths are several centimeters long (Table 7.2). The range in air, $\hat{R}_{air}$, for α-particles with an initial energy E_α MeV can be calculated by the empirical equation ($\rho_{air} = 1.293\ kg\ m^{-3}$; $1\ mg/cm^2 = 0.01\ kg/m^2$):

$$\hat{R}_{air} = 0.31\ \ {E_\alpha}^{3/2}(\text{cm}) = 0.40\ \ {E_\alpha}^{3/2}\left(\text{mg cm}^{-2}\right) = 0.004\ {E_\alpha}^{3/2}\left(\text{kg m}^{-2}\right) \quad (7.11)$$

The range $\hat{R}_z$ in other materials can be approximated roughly by

$$\hat{R}_z = 0.173\ \ {E_\alpha}^{3/2}\,{A_z}^{1/3}\left(\text{mg cm}^{-2}\right) = 0.00173\ {E_\alpha}^{3/2}\,{A_z}^{1/3}\left(\text{kg m}^{-2}\right) \quad (7.12)$$

where A_z is the atomic weight of the absorber. Figure 7.6 shows the range of various charged particles in an aluminum absorber. The range of a 5 MeV α is 6 mg cm^{-2}; thus

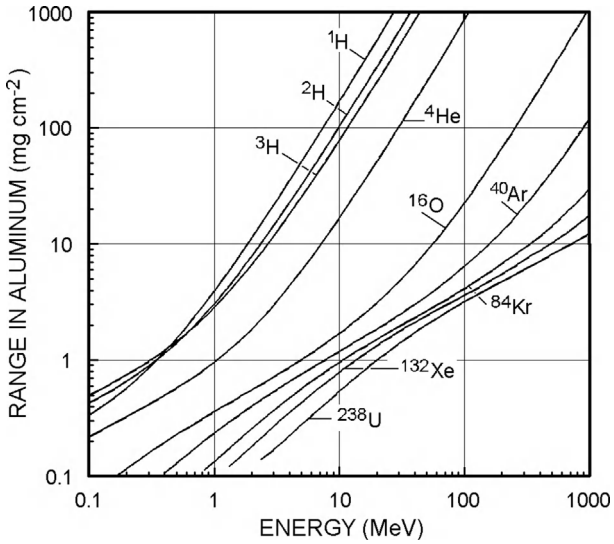


Figure 7.6 Range of some energetic particles in an aluminum absorber.

$\hat{R}_{\mathrm{Al}} = 6 \times 10^{-3}/\rho_{\mathrm{Al}}$ cm $= 0.002$ mm. Alpha-particles from radioactive decay are easily stopped even by a sheet of paper.

When the absorber consists of a composite material, containing the weight fractions w_1, w_2, w_3, etc of elements 1, 2, 3, etc with ranges $\hat{R}_1$, $\hat{R}_2$, $\hat{R}_3$, etc, the range $\hat{R}_{\mathrm{comp}}$ in the absorber is obtained from the relation

$$1/\hat{R}_{\mathrm{comp}} = w_1/\hat{R}_1 + w_2/\hat{R}_2 + w_3/\hat{R}_3 + \ldots \tag{7.13}$$

The number of ion pairs formed per millimeter of range for α-particles, protons, and electrons are shown in Figure 7.7(a). The larger *specific ionization* of the α-particles compared to the protons is related to the fact that the former are doubly charged. In general the specific ionization increases with the ionic charge of the particle for the same kinetic energy. Fission fragments that initially have very large energies also have very large ionic charges leading to quite high specific ionization in their absorption in matter; their range is 2–3 cm in air and 2–3 mg cm^{-2} in aluminum.

A quantum-mechanical and relativistic analysis of the interaction between a fast moving positive ion of atomic number z and the electrons in the absorber leads to the following expression for the energy loss per unit distance, $\mathrm{d}E/\mathrm{d}x$ (J/m), travelled in an absorber

$$\begin{aligned} -\mathrm{d}E/\mathrm{d}x = {} & \left\{\left(4\pi(\gamma z\mathbf{e})^2\mathbf{e}^2 NZ\right)/\left(\mathbf{m}_{\mathrm{e}}v^2\right)\right\}\left[\ln\left\{\left(2\mathbf{m}_{\mathrm{e}}v^2\right)/I\right\}\right. \\ & \left. -\ln\left(1-(v/\mathbf{c})^2\right)-(v/\mathbf{c})^2\right] \end{aligned} \tag{7.14}$$

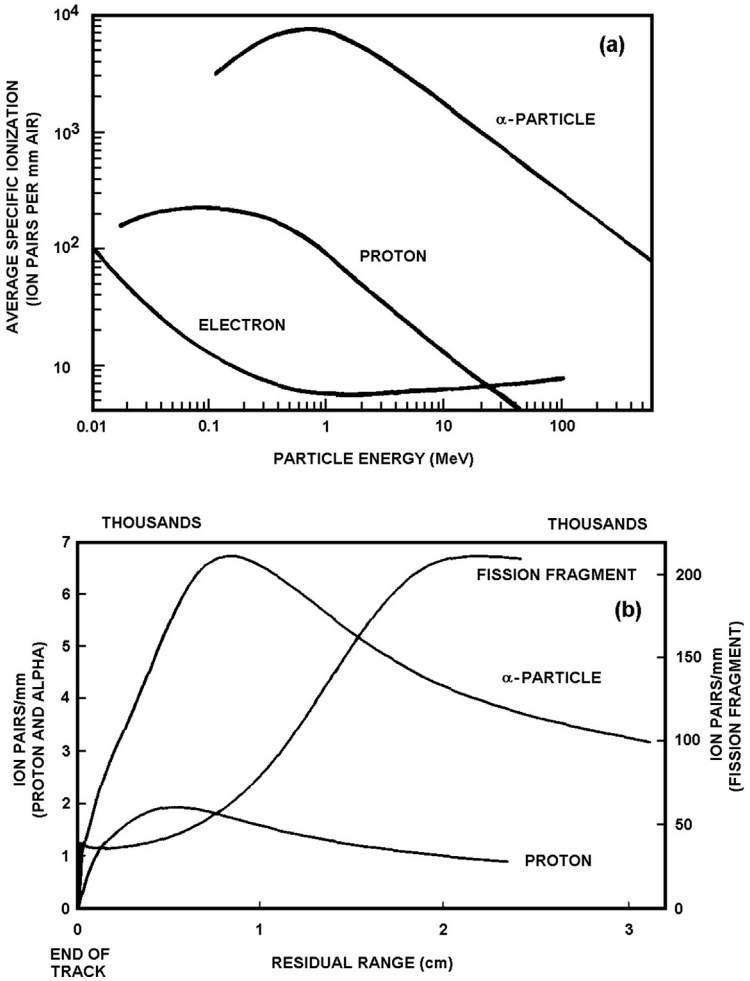


Figure 7.7 Number of ion pairs formed per mm air at STP as function of particle energy. (a) average specific ionization at maximum particle energy, (b) same for residual range. *(From H. A. C. McKay.)*

where $(\gamma z\mathbf{e})$ is the charge of the ion moving at velocity v through an absorber containing N atoms of atomic number Z per volume unit and having an effective ionization potential I. For a completely stripped ion, $\gamma = 1$. The range, $\hat{R}$, of an ion may be calculated by integrating the energy loss expression

$$\hat{R} = \int_{E_0}^{0} (\mathrm{d}E/\mathrm{d}x)^{-1}\mathrm{d}E \tag{7.15}$$

from the initial energy E_0 to zero.

Charged particles decrease in velocity as they lose their energy in traversing an absorber. As a result they spend progressively longer times in the vicinity of any particular atom, which results in an increase in the probability of interaction with that atom. Consequently there is a steady increase in the number of ion pairs formed along the path (or track) of the particle rather than a constant density of ion pairs. As mentioned above, near the end of the range for heavy charged particles a maximum is observed for the number of ion pairs formed per unit path length (the *Bragg peak*) (Fig. 7.7b). At a distance just beyond the Bragg peak maximum the kinetic energy of the particles is comparable to those of the orbital electrons of the absorber. As a result, the particle can acquire electrons, $\gamma < 1$ in Eqn. (7.14), finally becoming uncharged, $\gamma = 0$, and thereby losing its ability to cause further ionization. When scaled by the maximum ionization, the Bragg peak is most pronounced for protons.

7.4. ABSORPTION OF ELECTRONS

Absorption of high energy electron beams occurs through interaction with the orbital electrons and the electromagnetic field at the atom. The processes are summarized in Table 7.1 and Figure 7.8. In order to distinguish between electrons from accelerators and those from β-decay we refer to the latter as β-*particles*.

7.4.1. Ionization

Beta-particles lose their energy primarily by the same mechanism as α-particles (Fig. 7.7a and b); however, there are several important differences. Since the masses of the β-particles and of the orbital electrons are the same at non-relativistic velocities, the β-particles can lose a large fraction of their energy in a single collision. The β-particle undergoes a wide angle deflection in such collisions and consequently β-particles are scattered out of the beam path all along the length. The secondary electrons ionized from the atom have such high energies that they cause extensive secondary ionization which provides 70—80% of the total ionization in β-absorption processes (Fig. 7.5b). Approximately half of the total energy of the β-particle is lost by ionization and half by excitation. The track formed is further discussed in §8.2.

The specific ionization from a β-particle is much lower than that from a heavy ion as can be seen in Figure 7.7a. This is due to the fact that for the same initial energy, β-particles have much greater velocity than have α-particles or protons because their mass is very much smaller than the mass of the heavy particles. This greater velocity results in a correspondingly lower ionization due to the considerable shorter time available for interaction with the electrons of a specific atom and gives a much longer range to β-particles. The erratic path observed for β-particles in Figure 7.5(c) is a result of the large energy transfer and the resulting large deflections involved in the encounters with the orbital electrons. However, at very high energies the β-particles have straight